

HW 5

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Townsend: 5.2, 5.3, 5.5, 5.6

Townsend 5.2 (5 pts)

Express the total-spin $s = 1$ states of two spin- $\frac{1}{2}$ particles given in (5.30a) and (5.30c) in terms of the states $|+x, +x\rangle$, $|+x, -x\rangle$, $|-x, +x\rangle$, and $|-x, -x\rangle$.

5.30 A

$$|1, 1\rangle = |+z, +z\rangle$$

We have the projection operator to get four coefficients of superposition

$$|+z, +z\rangle = \alpha|+x, +x\rangle + \beta|+x, -x\rangle + \gamma|-x, +x\rangle + \delta|-x, -x\rangle$$

$$\langle +x|_z = \frac{1}{\sqrt{2}}\langle +z| + \frac{1}{\sqrt{2}}\langle -z|$$

$$\langle -x|_z = \frac{1}{\sqrt{2}}\langle +z| - \frac{1}{\sqrt{2}}\langle -z|$$

We can now do

$$\begin{aligned} |+z\rangle_x &= |+x\rangle\langle +x| + z\rangle + |-x\rangle\langle -x| + z\rangle \\ &= \frac{1}{\sqrt{2}}|+x\rangle + \frac{1}{\sqrt{2}}|-x\rangle \end{aligned}$$

Having two particles in the $|+z\rangle$ state gives us

$$|+z, +z\rangle = \frac{1}{2}|+x, +x\rangle + \frac{1}{2}|+x, -x\rangle + \frac{1}{2}|-x, +x\rangle + \frac{1}{2}|-x, -x\rangle$$

We can do the same for the other:

$$\begin{aligned} |1, -1\rangle &= |-z, -z\rangle \\ &= \frac{1}{2}|+x, +x\rangle - \frac{1}{2}|+x, -x\rangle - \frac{1}{2}|-x, +x\rangle + \frac{1}{2}|-x, -x\rangle \end{aligned}$$

Now we can find $\hat{S}^2$.

We know that

$$\begin{aligned} \hat{S}^2 &= (\hat{S}_1 + \hat{S}_2)^2 \\ &= \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_1\hat{S}_2 \\ \hat{S}_1^2|+x\rangle_1 &= \frac{1}{2}\left(\frac{1}{2} + 1\right)\hbar^2|+x\rangle \\ &= \frac{3}{4}\hbar^2|+x\rangle \\ \hat{S}_1^2|-x\rangle_1 &= \frac{-1}{2}\left(\frac{1}{2}\right)\hbar^2|-x\rangle \\ &= \frac{-\hbar^2}{4}|-x\rangle \end{aligned}$$

The same applies for $\hat{S}_2^2$.

We still have to find

$$2\hat{S}_1\hat{S}_2|+x, -x\rangle = 2\hat{S}_1\hat{S}_2|-x, +x\rangle = -\frac{3}{2}\hbar^2$$

$$2\hat{S}_1\hat{S}_2|+z, +z\rangle = 2\hat{S}_1\hat{S}_2|-z, -z\rangle = \frac{\hbar^2}{2}$$

old work

$$|+z\rangle = \frac{1}{\sqrt{2}}|+x\rangle + \frac{1}{\sqrt{2}}|-x\rangle$$

We know that

$$\hat{S}^2 = (\hat{S}_1 + \hat{S}_2)^2$$

$$\text{And } \hat{S}_1^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$$

Because each particle is in the $|+z\rangle$ state, we just have

$$\hat{S}_1^2 = \hat{S}_z^2$$

The same applies for 2.

$$\begin{aligned}\hat{S}^2|1, 1\rangle &= (\hat{S}_z^2 + \hat{S}_z^2 + \hat{S}_z\hat{S}_z)|+z, +z\rangle \\ &= 3\hat{S}_z^2|+z, +z\rangle \\ &= 3\left(\frac{1}{2}\left(\frac{1}{2} + 1\right)\right)\hbar^2|+z, +z\rangle \\ &= \frac{9}{4}\hbar^2|+z, +z\rangle\end{aligned}$$

Therefore,

$$\hat{S}|1, 1\rangle = \frac{3}{2}\hbar^2|+z, +z\rangle$$

Townsend 5.3 (5 pts)

Express the total-spin $s = 0$ state of two spin- $\frac{1}{2}$ particles given in (5.31) in terms of the states $|+n, -n\rangle$ and $|-n, +n\rangle$, where for a single spin- $\frac{1}{2}$ particle

$$\begin{aligned}|+n\rangle &= \cos\frac{\theta}{2}|+z\rangle + e^{i\phi}\sin\frac{\theta}{2}|-z\rangle \\ |-n\rangle &= \sin\frac{\theta}{2}|+z\rangle - e^{i\phi}\cos\frac{\theta}{2}|-z\rangle\end{aligned}$$

and

$$|0, 0\rangle = \frac{1}{\sqrt{2}}|+z, -z\rangle - \frac{1}{\sqrt{2}}|-z, +z\rangle \quad (5.31)$$

We can rearrange to find $|+z\rangle$ and $|-z\rangle$ in the n basis. This is just a system of equations.

$$\begin{aligned}\cos\frac{\theta}{2}|+n\rangle &= \cos^2\frac{\theta}{2}|+z\rangle + e^{i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}|-z\rangle \\ \sin\frac{\theta}{2}|-n\rangle &= \sin^2\frac{\theta}{2}|+z\rangle - e^{i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}|-z\rangle\end{aligned}$$

Adding these, we get

$$|+z\rangle = \cos\frac{\theta}{2}|+n\rangle + \sin\frac{\theta}{2}|-n\rangle$$

We can do the same for $|-z\rangle$, just subtracting

$$\begin{aligned}\sin \frac{\theta}{2} | +n \rangle &= \cos \frac{\theta}{2} \sin \frac{\theta}{2} | +z \rangle + e^{i\phi} \sin^2 \frac{\theta}{2} | -z \rangle \\ \cos \frac{\theta}{2} | -n \rangle &= \cos \frac{\theta}{2} \sin \frac{\theta}{2} | +z \rangle - e^{i\phi} \cos^2 \frac{\theta}{2} | -z \rangle \\ e^{i\phi} | -z \rangle &= \sin \frac{\theta}{2} | +n \rangle - \cos \frac{\theta}{2} | -n \rangle \\ | -z \rangle &= e^{-i\phi} \sin \frac{\theta}{2} | +n \rangle - e^{-i\phi} \cos \frac{\theta}{2} | -n \rangle\end{aligned}$$

and

$$|0, 0\rangle = \frac{1}{\sqrt{2}} | +\mathbf{z}, -\mathbf{z} \rangle - \frac{1}{\sqrt{2}} | -\mathbf{z}, +\mathbf{z} \rangle \quad (5.31)$$

This gives us

$$\begin{aligned}| +z, -z \rangle &= \left(\cos \frac{\theta}{2} | +n \rangle + \sin \frac{\theta}{2} | -n \rangle \right) \left(e^{-i\phi} \sin \frac{\theta}{2} | +n \rangle - e^{-i\phi} \cos \frac{\theta}{2} | -n \rangle \right) \\ &= e^{-i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} | +n, +n \rangle - e^{-i\phi} \cos^2 \frac{\theta}{2} | +n, -n \rangle + e^{-i\phi} \sin^2 \frac{\theta}{2} | -n, +n \rangle - \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{-i\phi} | -n, -n \rangle\end{aligned}$$

We can also do this for

$$\begin{aligned}| -z, +z \rangle &= \left(e^{-i\phi} \sin \frac{\theta}{2} | +n \rangle - e^{-i\phi} \cos \frac{\theta}{2} | -n \rangle \right) \left(\cos \frac{\theta}{2} | +n \rangle + \sin \frac{\theta}{2} | -n \rangle \right) \\ &= e^{-i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} | +n, +n \rangle + e^{-i\phi} \sin^2 \frac{\theta}{2} | +n, -n \rangle - e^{-i\phi} \cos^2 \frac{\theta}{2} | -n, +n \rangle - e^{-i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} | -n, -n \rangle\end{aligned}$$

(SIDE NOTE: TENSOR PRODUCTS DON'T COMMUTE BECAUSE ORDER OF N IN KETS MATTERS OOOOH)

We can add these two. The identical $| +n, +n \rangle$ and $| -n, -n \rangle$ cancel.

Finally, we get:

$$\begin{aligned}|0, 0\rangle &= \frac{1}{\sqrt{2}} \left(-e^{-i\phi} \cos^2 \frac{\theta}{2} - e^{-i\phi} \sin^2 \frac{\theta}{2} \right) | +n, -n \rangle + \frac{1}{\sqrt{2}} \left(+e^{-i\phi} \cos^2 \frac{\theta}{2} + e^{-i\phi} \sin^2 \frac{\theta}{2} \right) | -n, +n \rangle \\ &= \frac{-1}{\sqrt{2}} e^{-i\phi} | +n, -n \rangle + \frac{1}{\sqrt{2}} e^{-i\phi} | -n, +n \rangle\end{aligned}$$

Townsend 5.5 (5 pts)

At time $t = 0$, an electron and a positron are formed in a state with total spin angular momentum equal to zero, perhaps from the decay of a spinless particle. The particles are situated in a uniform magnetic field B_0 in the z direction.

- (a) If interaction between the electron and the positron may be neglected, show that the spin Hamiltonian of the system may be written as

$$\hat{H} = \omega_0 (\hat{S}_{1z} - \hat{S}_{2z})$$

where $\hat{S}_1$ is the spin operator of the electron, $\hat{S}_2$ is the spin operator of the positron, and ω_0 is a constant.

- (b) What is the spin state of the system at time t ? Show that the state of the system oscillates between a spin-0 and a spin-1 state. Determine the period of oscillation.
(c) At time t , measurements are made of S_{1z} and S_{2z} . Calculate the probability that *both* of these measurements yield the value $\hbar/2$.

and

$$|0, 0\rangle = \frac{1}{\sqrt{2}} | +\mathbf{z}, -\mathbf{z} \rangle - \frac{1}{\sqrt{2}} | -\mathbf{z}, +\mathbf{z} \rangle \quad (5.31)$$

We have

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} | +z, -z \rangle - \frac{1}{\sqrt{2}} | -z, +z \rangle$$

The Spin Hamiltonian for an electron is

$$\hat{\mu}_z = -\frac{ge}{2mc} \hat{S}_z$$

For a positron, this is

$$\hat{\mu}_z = \frac{ge}{2mc} \hat{S}_z$$

The Hamiltonian for each is

$$\hat{H} = -\hat{\mu}_z B$$

The difference between these is

$$H_e - H_p = \frac{-ge}{2mc} (\hat{S}_{ez} - \hat{S}_{pz})$$

If we label

$$\omega_0 = -\frac{ge}{2mc}$$

then we have

$$\hat{H} = \omega_0 (\hat{S}_{1z} - \hat{S}_{2z})$$

b

(b) What is the spin state of the system at time t ? Show that the state of the system oscillates between a spin-0 and a spin-1 state. Determine the period of oscillation.

We can find the eigenvalues of the operators $\hat{S}_{1,z}$ and $\hat{S}_{2,z}$.

$$\begin{aligned}\hat{S}|z\rangle &= \frac{\hbar}{2}|z\rangle \\ \hat{S}|-z\rangle &= -\frac{\hbar}{2}|-z\rangle\end{aligned}$$

The spin zero states are $|+z, -z\rangle$ and $|-z, +z\rangle$, while the spin 1 states are $|-z, -z\rangle$ and $|+z, +z\rangle$.

$$\begin{aligned}\hat{H}|+z, -z\rangle &= \omega_0 \hbar | +z, -z\rangle \\ \hat{H}|-z, +z\rangle &= -\omega_0 \hbar | -z, +z\rangle\end{aligned}$$

We can find the Hamiltonian acting on each z_1, z_2 - as they rotate they'll align and move away, shifting between these four states.

If we start in spin zero, then we have the time evolution given by

$$\psi(t) = e^{-i\omega_0 t} | +z\rangle_1 \otimes e^{i\omega_0 t} | -z\rangle_2$$

(side note to the gruter - do we have to normalize this with a $\frac{1}{\sqrt{2}}$ I'm not sure, since each one is definitely in that state and not split, so each has magnitude 1? It feels like for problem c i need a normalization though)

These are 90 degrees apart and counter rotating. When one has moved π , the other has moved $-\pi$, so they are aligned into a spin 1 state. Another π and they are in opposite spins, and in a spin zero state.

This gives the period of oscillation as

$$t = \frac{\pi}{\omega_0}$$

C

(c) At time t , measurements are made of S_{1x} and S_{2x} . Calculate the probability that *both* of these measurements yield the value $\hbar/2$.

$$\psi_1(t) = e^{-\frac{i}{2}t}|+z\rangle$$

We can find this as

$$(\langle +x_1 | \psi(t)_1 \rangle)^2 + (\langle +x_2 | \psi(t)_2 \rangle)^2$$

We know that

$$\begin{aligned}\langle +x | +z \rangle &= \frac{1}{2} \\ \langle +x | -z \rangle &= \frac{1}{2}\end{aligned}$$

For the first term, we have

$$\langle +x | e^{i\omega_0 t} | +z \rangle$$

old work

We can write out our time evolving states as

$$\begin{aligned}\psi_1(t) &= e^{-\frac{i}{2}t}|+z\rangle \\ |+z\rangle &= \psi(t)e^{\frac{i}{2}t} \\ |+x\rangle &= \frac{1}{\sqrt{2}}|+z\rangle + \frac{1}{\sqrt{2}}|-z\rangle \\ &= \frac{1}{\sqrt{2}}e^{\frac{i}{2}t}|\psi_1(t)\rangle +\end{aligned}$$

We will measure $\frac{\hbar}{2}$ half of the time in a pure $|+z\rangle$ and a pure $|-z\rangle$.

Townsend 5.6 (5 pts)
 Take the spin Hamiltonian of the positronium atom (a bound state of an electron and a positron) in an external magnetic field in the z direction to be

$$\hat{H} = \frac{2A}{\hbar^2} \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 + \omega_0 (\hat{S}_{1z} - \hat{S}_{2z})$$

Determine the energy eigenvalues.

$$\hat{S}_1 \cdot \hat{S}_2 = \hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z}$$

$$\begin{aligned}S_+ &= S_x + iS_y \\ S_- &= S_x - iS_y\end{aligned}$$

We can rewrite this with the raising and lowering operators

$$\begin{aligned}S_x &= \frac{S_+ + S_-}{2} \\ S_y &= \frac{S_+ - S_-}{2i}\end{aligned}$$

We can write out the spin $\otimes$ product with the raising and lowering operators as

$$\begin{aligned}\hat{S}_1 \cdot \hat{S}_2 &= \hat{S}_{1z}\hat{S}_{2z} + \frac{\hat{S}_{1+} + \hat{S}_{1-}}{2} \frac{\hat{S}_{2+} + \hat{S}_{2-}}{2} + \frac{\hat{S}_{1+} - \hat{S}_{1-}}{2i} \frac{\hat{S}_{2+} - \hat{S}_{2-}}{2i} \\ &= \hat{S}_{1z}\hat{S}_{2z} + \frac{1}{4}(S_{1+}S_{2+} + S_{1+}S_{2-} + S_{1-}S_{2+} + S_{1-}S_{2-}) - \frac{1}{4}(S_{1+}S_{2+} + S_{1-}S_{2-}) \\ &= \hat{S}_{1z}\hat{S}_{2z} + \frac{1}{4}(S_{1+}S_{2-} + S_{1-}S_{2+})\end{aligned}$$

Also, note that

$$\begin{aligned}\hat{S}_+|+z\rangle &= 0 \\ \hat{S}_+|-z\rangle &= \hbar|+z\rangle \\ \hat{S}_-|+z\rangle &= -\hbar|-z\rangle \\ \hat{S}_-|-z\rangle &= 0\end{aligned}$$

We can now find the effect of the Hamiltonian on each state

$$\begin{aligned}\hat{H}|+z, +z\rangle &= \frac{2A}{\hbar^2} \left(\frac{\hbar^2}{4} |+z, +z\rangle \right) \\ &= \frac{A}{2} |+z, +z\rangle \\ \hat{H}|+z, -z\rangle &= \frac{2A}{\hbar^2} \left(-\frac{\hbar^2}{4} |+z, -z\rangle - \frac{\hbar^2}{2} |-z, +z\rangle \right) + \omega_0 \hbar \\ \hat{H}|-z, +z\rangle &= \frac{2A}{\hbar^2} \left(\frac{\hbar^2}{4} |-z, +z\rangle - \frac{\hbar^2}{2} |+z, -z\rangle \right) + \omega_0 \hbar \\ \hat{H}|-z, -z\rangle &= \frac{2A}{\hbar^2} \left(\frac{\hbar^2}{4} |-z, -z\rangle \right) \\ &= \frac{A}{2} |-z, -z\rangle\end{aligned}$$

This gives us the matrix

$$\hat{H} = \begin{bmatrix} \frac{A}{2} & 0 & 0 & 0 \\ 0 & -\frac{A}{2} + \omega_0 \hbar & -A & 0 \\ 0 & -A & \frac{A}{2} + \omega_0 \hbar & 0 \\ 0 & 0 & 0 & A \end{bmatrix}$$

We can find the eigenvalues for the innermost matrix

$$\begin{bmatrix} -\frac{A}{2} + \omega_0 \hbar & A \\ A & -\frac{A}{2} - \omega_0 \hbar \end{bmatrix}$$

This gets

$$\begin{aligned}\left(-\frac{A}{2} + \omega_0 \hbar - \lambda \right) \left(-\frac{A}{2} - \omega_0 \hbar - \lambda \right) - A^2 &= 0 \\ \left(\left(-\frac{A}{2} - \lambda \right)^2 - \omega^2 \hbar^2 \right) - A^2 &= 0 \\ \left(-\frac{A}{2} - \lambda \right)^2 &= A^2 + \omega^2 \hbar^2 \\ -\frac{A}{2} - \lambda &= \pm \sqrt{A^2 + \omega^2 \hbar^2} \\ \lambda &= - \left(\frac{A}{2} \pm \sqrt{A^2 + \omega^2 \hbar^2} \right)\end{aligned}$$

The top left and bottom right entry don't multiply with anything

Our eigenvalues are thus

$$\frac{A}{2}, - \left(\frac{A}{2} + \sqrt{A^2 + \omega^2 \hbar^2} \right), - \left(\frac{A}{2} - \sqrt{A^2 + \omega^2 \hbar^2} \right), \frac{A}{2}$$

old work, thanks lan for helping

We can check the effect of the Hamiltonian on each of the spin $\frac{1}{2}$ states.

$$\begin{aligned}\hat{H}|+z, +z\rangle &= \left(\frac{2A}{\hbar^2} \left(\frac{1}{2} \frac{1}{2} \hbar^2 \right) + \omega_0 \hbar(0) \right) |+z, +z\rangle \\ \hat{H}|+z, -z\rangle &= \left(\frac{2A}{\hbar^2} \left(\frac{1}{2} \frac{-1}{2} \right) \hbar^2 + \omega_0 \hbar(1) \right) |+z, -z\rangle \\ \hat{H}|-z, +z\rangle &= \left(\frac{2A}{\hbar^2} \left(\frac{1}{2} \frac{-1}{2} \right) \hbar^2 + \omega_0 \hbar(0) \right) |-z, +z\rangle \\ \hat{H}|-z, -z\rangle &= \left(\frac{2A}{\hbar^2} \left(\frac{-1}{2} \frac{-1}{2} \right) \hbar^2 + \omega_0 \hbar(0) \right) |-z, -z\rangle\end{aligned}$$

These simplify down to

$$\begin{aligned}\hat{H}|+z, +z\rangle &= \frac{A}{2} |+z, +z\rangle \\ \hat{H}|+z, -z\rangle &= -\left(\frac{A}{2} + \omega_0 \right) |+z, -z\rangle \\ \hat{H}|-z, +z\rangle &= -\left(\frac{A}{2} + \omega_0 \right) |-z, +z\rangle \\ \hat{H}|-z, -z\rangle &= -\frac{A}{2} |-z, -z\rangle\end{aligned}$$

We can write this in a matrix

$$\hat{H} = \begin{bmatrix} \frac{A}{2} & 0 & 0 & 0 \\ 0 & -\frac{A}{2} + \omega_0 \hbar & 0 & 0 \\ 0 & 0 & -\frac{A}{2} + \omega_0 \hbar & 0 \\ 0 & 0 & 0 & -\frac{A}{2} \end{bmatrix}$$

This is bad, the $|+z, +z\rangle$ is not the eigenbasis of the hamiltonian, I can't just construct stuff like this. RAGH